

Eventually Monotone Variable-Exponent Iterated L_p Spaces

Run fa_banach_001, agent_lane_13

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Status

This is a `partial_result_likely_valid` packet for the variable exponent problem in Yanqi Qiu, *On the UMD constants for a class of iterated $L_p(L_q)$ spaces*, arXiv:1112.0739. The result gives a positive answer for eventually monotone exponent sequences. It is not a full characterization.

Source Problem

Qiu defines, for $1 < p_i < \infty$,

$$X[(p_i)] = \varinjlim L_{p_n} \cdots L_{p_2} L_{p_1}, \quad Z[(p_i)] = \varinjlim L_{p_1} L_{p_2} \cdots L_{p_n},$$

and asks:

Under which condition is $X[(p_i)]$ or $Z[(p_i)]$ in the UMD class?

The paper records necessary conditions: the exponents must remain in a compact subinterval of $(1, \infty)$, must have a unique cluster point, and must pass the product obstruction

$$\prod_i c(p_i, p_{i+1}) < \infty.$$

It then says that, intuitively, sufficiently fast convergence $p_i \rightarrow p$ should imply the UMD property.

Problem. *Under which condition is $X[(p_i)]$ or $Z[(p_i)]$ in the UMD class ?*

We have the following observations on the necessary condition:

- (i) A trivial necessary condition is that there exist $1 < p_0, p_\infty < \infty$, such that $p_0 \leq p_i \leq p_\infty$ for all $i \geq 1$.
- (ii) If the above condition is satisfied, then the sequence (p_i) has at least one cluster point $1 < p < \infty$. Then a necessary condition is that the sequence has only one cluster point, i.e. $\lim_{i \rightarrow \infty} p_i = p$. Indeed, assume that the sequence (p_i) has two cluster points $1 < p \neq q < \infty$, so that there exist two subsequences of (p_i) which tend to p, q respectively. Then one can easily show that by Theorem 3.10, both $X[(p_i)]$ and $Z[(p_i)]$ are non-UMD (they are in fact non-AUMD).
- (iii) Now the speed of convergence of (p_i) will play a role. Since $\ell_{p_1}^{(2)}(\ell_{p_2}^{(2)}(\dots(\ell_{p_n}^{(2)})\dots))$ embeds isometrically into $L_{p_1}L_{p_2}\dots L_{p_n}$. A necessary condition for $Z[(p_i)]$ to be UMD is $\prod_i c(p_{2i}, p_{2i+1}) < \infty$. Similarly, it is necessary that $\prod_i c(p_{2i+1}, p_{2i+2}) < \infty$. Combining these, a necessary condition for $Z[(p_i)]$ to be in the UMD class is

$$\prod_i c(p_i, p_{i+1}) < \infty.$$

The same statement remains true for $X[(p_i)]$. Note that by (4), $c(p_i, p_{i+1}) > 1$ if $p_i \neq p_{i+1}$.

Intuitively, if p_i tends to p sufficiently fast, then both $X[(p_i)]$ and $Z[(p_i)]$ are in the UMD class. The author obtained some partial results in this direction, which will be treated elsewhere.

Idea of the Proof

Qiu's Remark 5.7 contains a finite-dimensional monotonicity observation: a consecutive monotone block in a finite mixed-norm tower may be compressed to its two endpoints without changing the UMD_s constant. Therefore, if the tail of (p_i) is monotone, the whole tail $p_N, \dots, p_n$ contributes no more than the two exponents p_N and p_n at level n . Since p_n stays inside one compact interval and all other exponents left after compression are fixed, the finite-level UMD_s constants are uniformly bounded. Uniform finite-level bounds pass to the inductive limit.

in this direction, which will be treated elsewhere.

Remark 5.7. Let $1 < p < q < \infty$. We have the following Banach lattices isometries

$$L_p L_q = L_p L_p L_q, \quad L_p L_q = L_p L_q L_q.$$

Since $L_p L_r L_q$ is an interpolation space between $L_p L_p L_q$ and $L_p L_q L_q$ for any $p \leq r \leq q$, the UMD_s constant of $L_p L_r L_q$ is actually the same as that of $L_p(L_q)$. The same argument shows that $L_p L_u L_r L_v L_q$ has the same UMD_s constant with $L_p L_q$, provided $p \leq u \leq r \leq v \leq q$. More generally, if $(p_i)_{i=1}^n$ is a finite sequence, assume that $(p_i)_{i=k}^l$ is consecutive monotone (non-increasing or non-decreasing) subsequence, then $L_{p_1} \cdots L_{p_k} \cdots L_{p_l} \cdots L_{p_n}$ and $L_{p_1} \cdots L_{p_k} L_{p_l} \cdots L_{p_n}$ have the same UMD_s constant for all $1 < s < \infty$.

A Standard Stability Lemma

Lemma 1. Fix $1 < s < \infty$ and a compact interval $I = [a, b] \subset (1, \infty)$. There is a constant $K = K(I, s)$ such that

$$C_s(L_r(Y)) \leq K C_s(Y)$$

for every $r \in I$ and every UMD Banach space Y , where C_s denotes the UMD_s martingale transform constant.

Proof. This is the usual vector-valued L_r -stability of the UMD property, with constants locally bounded for r in compact subintervals of $(1, \infty)$. Equivalently, one may combine the standard independence of the UMD exponent, used in Qiu's equation (10), with the identity $C_r(L_r(Y)) = C_r(Y)$. The resulting constants are bounded uniformly for $r \in I$. $\square$

Partial Theorem

Theorem 1. Let $(p_i)_{i \geq 1}$ satisfy

$$1 < a \leq p_i \leq b < \infty$$

for all i . Suppose that (p_i) is monotone from some index N onward. Then both inductive limits $X[(p_i)]$ and $Z[(p_i)]$, in Qiu's notation, are UMD.

Proof. Fix $1 < s < \infty$. It is enough to show that the UMD_s constants of the finite levels are bounded uniformly in the level.

Consider first

$$Z_n = L_{p_1} L_{p_2} \cdots L_{p_n}.$$

For $n \geq N$, the block $p_N, p_{N+1}, \dots, p_n$ is consecutive and monotone. By Qiu's Remark 5.7, deleting the interior of this block does not change the UMD_s constant. Hence

$$C_s(Z_n) = C_s(L_{p_1} \cdots L_{p_N} L_{p_n}) \quad (n \geq N).$$

The right-hand side contains only the fixed layers $p_1, \dots, p_N$ and one variable layer $p_n \in [a, b]$. Applying Lemma 1 a finite number of times gives

$$\sup_{n \geq N} C_s(Z_n) < \infty.$$

The finitely many levels $n < N$ do not affect boundedness.

For

$$X_n = L_{p_n} L_{p_{n-1}} \cdots L_{p_1},$$

the reversed tail $p_n, p_{n-1}, \dots, p_N$ is again consecutive and monotone. Qiu's same remark gives

$$C_s(X_n) = C_s(L_{p_n} L_{p_{n-1}} \cdots L_{p_1}) \quad (n \geq N).$$

Again only one exponent, p_n , varies in the compact interval $[a, b]$, while all other layers are fixed. Lemma 1 yields

$$\sup_{n \geq N} C_s(X_n) < \infty.$$

Finally, the inductive limits are the closures of increasing unions of the finite levels. Any finite martingale transform test with values in the dense algebraic union takes its values in some finite level, where the same uniform constant applies. A density argument passes the inequality to the completed inductive limit. Thus both $X[(p_i)]$ and $Z[(p_i)]$ are UMD. $\square$

Limitations

The theorem does not characterize all exponent sequences. It does not settle oscillatory sequences with infinitely many local extrema, nor does it prove that Qiu's necessary product condition $\prod_i c(p_i, p_{i+1}) < \infty$ is sufficient. It also does not compute the constants $c(p, q)$, which Qiu explicitly notes are unknown in general.

Search and Novelty Check

The lane-13 queue selected arXiv:1112.0739. Cheap run indexes were searched for the arXiv id, title, UMD constants, $X[(p_i)]$, $Z[(p_i)]$, and the product condition; no duplicate packet or attempt was found. Local source search found the exact variable-exponent problem only in arXiv:1112.0739. Targeted web searches for exact phrases including `Under which condition is  $X[(p_i)]$ ,  $Z[(p_i)]$  UMD, and  $\prod_i c(p_i, p_{i+1}) < \infty$` found no later exact answer beyond the source paper. This packet should therefore be read as a source-derived partial theorem, not as a known later-literature resolution.

Human Review Recommendation

The main point to check is the use of Qiu's Remark 5.7 for the reversed finite towers defining X_n . Since reversing an eventually monotone block keeps it monotone, the same deletion statement applies, but this is the one step where notation and tower order should be verified against the source.

References

- [1] Y. Qiu, *On the UMD constants for a class of iterated $L_p(L_q)$ spaces*, arXiv:1112.0739.